

Math 1512 Calculus 1 Review: Selected Solutions

Fall 2026

One worked problem from each topic. Problem numbers match the review sheet, and each problem is restated before its solution.

1. Formulas to know.

- (c) Give both parts of the Fundamental Theorem of Calculus.

Solution. Let f be continuous on $[a, b]$.

Part 1. If $g(x) = \int_a^x f(t) dt$, then $g'(x) = f(x)$. The rate the area grows is the height of the graph at the moving edge.

Part 2. If F is any antiderivative of f , meaning $F' = f$, then $\int_a^b f(x) dx = F(b) - F(a)$. Adding up tiny changes in F gives the total change in F . ■

2. Limits from a graph.

- (b) At which x values in $(-3, 4)$ is g not continuous? Classify each discontinuity. At which of these is g continuous from the left, or from the right?

Solution. g is not continuous at $x = -1, 0, 2$.

At $x = -1$ the one-sided limits are 3 and 4: a jump. At $x = 0$ the limit exists ($= 1$) but is not $g(0) = 3$: removable, since redefining one point fixes it. At $x = 2$ the right-hand limit is infinite: an infinite discontinuity.

Continuous from the left at a means $\lim_{x \rightarrow a^-} g(x) = g(a)$. This holds only at $x = 2$, where both equal 1. No point is continuous from the right: at -1 we compare 4 with 1, at 0 we compare 1 with 3, and at 2 the right limit is infinite. ■

3. Computing limits.

- (c) Evaluate $\lim_{x \rightarrow -1} \frac{\frac{1}{x+3} - \frac{1}{2}}{x+1}$.

Solution. Combine the fractions in the numerator over the common denominator $2(x+3)$.

$$\frac{1}{x+1} \cdot \frac{2 - (x+3)}{2(x+3)} = \frac{1}{x+1} \cdot \frac{-(x+1)}{2(x+3)} = \frac{-1}{2(x+3)} \longrightarrow -\frac{1}{4}.$$

■

4. Squeeze Theorem.

- (a) Evaluate $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{3}{x}\right)$.

Solution. $\sin\left(\frac{3}{x}\right)$ oscillates and has no limit, so the product law does not apply (this answers (c)). But it is bounded: $-1 \leq \sin\left(\frac{3}{x}\right) \leq 1$. Multiplying by $x^2 \geq 0$,

$$-x^2 \leq x^2 \sin\left(\frac{3}{x}\right) \leq x^2.$$

Both bounds go to 0, so by the Squeeze Theorem the limit is 0. The x^2 crushes the oscillation. ■

5. Limits at infinity.

- (a) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2+1}}{x+3}$.

Solution. Pull x^2 out of the root: $\sqrt{4x^2+1} = \sqrt{x^2} \sqrt{4 + \frac{1}{x^2}} = |x| \sqrt{4 + \frac{1}{x^2}}$. For $x < 0$, $|x| = -x$.

$$\frac{-x \sqrt{4 + \frac{1}{x^2}}}{x \left(1 + \frac{3}{x}\right)} = \frac{-\sqrt{4 + \frac{1}{x^2}}}{1 + \frac{3}{x}} \rightarrow -2.$$

The sign makes sense: the top is positive and the bottom is negative for very negative x . ■

6. Continuity and the IVT.

- (b) Find a and b so that $f(x) = \begin{cases} 2 & x < 1 \\ ax + b & 1 \leq x < 4 \\ 8 & x \geq 4 \end{cases}$ is continuous everywhere.

Solution. Each piece is continuous, so only the seams at $x = 1$ and $x = 4$ matter. The middle line must start at height 2 and end at height 8.

$$a + b = 2, \quad 4a + b = 8.$$

Subtracting, $3a = 6$, so $a = 2$ and $b = 0$. The middle piece is $2x$. ■

7. The derivative as a limit.

- (b) Compute $f'(x)$ directly from $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ for $f(x) = \frac{x}{x+2}$.

Solution. Combine the fractions in the numerator first.

$$\frac{1}{h} \left(\frac{x+h}{x+h+2} - \frac{x}{x+2} \right) = \frac{(x+h)(x+2) - x(x+h+2)}{h(x+h+2)(x+2)} = \frac{2h}{h(x+h+2)(x+2)}.$$

Cancel h and let $h \rightarrow 0$: $f'(x) = \frac{2}{(x+2)^2}$. ■

8. Differentiation rules.

- (b) Find the points on the curve $y = x^3 - 3x^2 - 9x + 2$ where the tangent line is parallel to the line $y = 15x$.

Solution. Parallel lines have equal slopes, so we need $y' = 15$.

$$3x^2 - 6x - 9 = 15 \implies x^2 - 2x - 8 = 0 \implies (x-4)(x+2) = 0.$$

At $x = 4$, $y = 64 - 48 - 36 + 2 = -18$, and at $x = -2$, $y = -8 - 12 + 18 + 2 = 0$. The points are $(4, -18)$ and $(-2, 0)$. ■

9. Trig derivatives.

- (b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$.

Solution.

$$\frac{\sin(4x)}{3x} = \frac{4}{3} \cdot \frac{\sin(4x)}{4x} \longrightarrow \frac{4}{3}.$$

For the second, multiply by $\frac{1+\cos x}{1+\cos x}$ and use $1 - \cos^2 x = \sin^2 x$.

$$\frac{1 - \cos^2 x}{x \sin x (1 + \cos x)} = \frac{\sin^2 x}{x \sin x (1 + \cos x)} = \frac{\sin x}{x} \cdot \frac{1}{1 + \cos x} \longrightarrow 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

■

10. Chain rule.

- (a) Use the table to find $(f \circ g)'(1)$, $(g \circ f)'(1)$, and $(fg)'(2)$.

x	0	1	2	3
$f(x)$	2	3	0	1
$f'(x)$	-1	4	2	5
$g(x)$	1	2	3	0
$g'(x)$	3	-2	1	4

Solution. The chain rule evaluates the outer derivative at the inner value:

$$(f \circ g)'(1) = f'(g(1)) g'(1) = f'(2) \cdot (-2) = 2(-2) = -4.$$

$$(g \circ f)'(1) = g'(f(1)) f'(1) = g'(3) \cdot 4 = 16.$$

$$\text{The product rule gives } (fg)'(2) = f'(2)g(2) + f(2)g'(2) = 2 \cdot 3 + 0 \cdot 1 = 6.$$

■

11. Implicit differentiation.

- (b) Find the tangent line to $\sin(x + y) = 2x - 2y$ at (π, π) .

Solution. Check the point: $\sin(2\pi) = 0 = 2\pi - 2\pi$. Differentiate.

$$\cos(x + y) (1 + y') = 2 - 2y'.$$

At the point $\cos(2\pi) = 1$, so $1 + y' = 2 - 2y'$ and $y' = \frac{1}{3}$. The tangent line is $y = \pi + \frac{1}{3}(x - \pi)$.

■

12. Related rates.

- (b) A tank is an inverted cone with radius 3 m at the top and height 6 m. Water is pumped in at 4 m³/min. How fast is the water level rising when the water is 4 m deep?

Solution. Let h be the depth and r the radius of the water surface. Similar triangles give $\frac{r}{h} = \frac{3}{6}$, so $r = \frac{h}{2}$ and

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{12}, \quad \frac{dV}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}.$$

At $h = 4$: $4 = 4\pi \frac{dh}{dt}$, so $\frac{dh}{dt} = \frac{1}{\pi}$ m/min. The factor $\frac{\pi h^2}{4}$ is the area of the water surface, which is why the level rises more slowly as the tank fills.

■

13. Linear approximation.

- (a) Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 27$. Use it to approximate $\sqrt[3]{27.27}$.

Solution. $f(27) = 3$ and $f'(x) = \frac{1}{3}x^{-2/3}$, so $f'(27) = \frac{1}{27}$. Then $L(x) = 3 + \frac{x-27}{27}$ and $L(27.27) = 3 + \frac{0.27}{27} = 3.01$. The true value is $3.00997\dots$ ■

14. Extreme values.

- (c) Find the absolute max and min of $f(x) = x\sqrt{4-x^2}$ on $[-1, 2]$.

Solution. By the product and chain rules,

$$f'(x) = \sqrt{4-x^2} - \frac{x^2}{\sqrt{4-x^2}} = \frac{4-2x^2}{\sqrt{4-x^2}}.$$

$f' = 0$ at $x = \pm\sqrt{2}$, and only $\sqrt{2}$ is in $[-1, 2]$. f' is undefined at $x = 2$, an endpoint anyway.

$f(-1) = -\sqrt{3}$, $f(\sqrt{2}) = \sqrt{2} \cdot \sqrt{2} = 2$, $f(2) = 0$. Max 2 at $x = \sqrt{2}$, min $-\sqrt{3}$ at $x = -1$. ■

15. Mean Value Theorem.

- (c) Show that $x^3 + 5x - 2 = 0$ has exactly one real solution.

Solution. Let $f(x) = x^3 + 5x - 2$. At least one: $f(0) = -2 < 0 < 4 = f(1)$, so the IVT gives a root in $(0, 1)$.

At most one: if $f(a) = f(b) = 0$ with $a < b$, the MVT gives c with $f'(c) = \frac{0-0}{b-a} = 0$. But $f'(x) = 3x^2 + 5 \geq 5$. So a second root is impossible. ■

16. Shape of a graph.

- (a) For $f(x) = x^3 - 6x^2 + 9x + 1$, find the intervals of increase and decrease, local extrema, concavity, and inflection points.

Solution. $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. So $f' > 0$ on $(-\infty, 1)$ and $(3, \infty)$ (increasing) and $f' < 0$ on $(1, 3)$ (decreasing).

f' changes from $+$ to $-$ at 1: local max $f(1) = 5$. From $-$ to $+$ at 3: local min $f(3) = 1$.

$f''(x) = 6x - 12$: concave down on $(-\infty, 2)$, concave up on $(2, \infty)$, inflection point $(2, 3)$. The inflection point sits halfway between the max and min, as for every cubic. ■

17. Curve sketching.

- (b) Sketch $f(x) = \frac{x^2}{x-1}$, labeling all key features.

Solution. Domain $x \neq 1$. The only intercept is $(0, 0)$. At $x = 1$ the top is $1 \neq 0$, so $x = 1$ is a vertical asymptote: $f \rightarrow -\infty$ as $x \rightarrow 1^-$ and $f \rightarrow \infty$ as $x \rightarrow 1^+$.

There is no horizontal asymptote, since the top has higher degree. Long division gives $\frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}$, and $\frac{1}{x-1} \rightarrow 0$, so the graph approaches the slant asymptote $y = x + 1$ as $x \rightarrow \pm\infty$.

$$f'(x) = \frac{x(x-2)}{(x-1)^2}, \quad f''(x) = \frac{2}{(x-1)^3}.$$

Increasing on $(-\infty, 0)$ and $(2, \infty)$, decreasing on $(0, 1)$ and $(1, 2)$. Local max $(0, 0)$, local min $(2, 4)$. Concave down on $(-\infty, 1)$, concave up on $(1, \infty)$, no inflection point. The graph is two branches hugging $y = x + 1$, one with a peak at the origin and one with a valley at $(2, 4)$. ■

18. Optimization.

- (b) A closed cylindrical can holds $16\pi \text{ cm}^3$. Find the radius and height that minimize its surface area.

Solution. $\pi r^2 h = 16\pi$, so $h = \frac{16}{r^2}$. The surface area is two disks plus the side.

$$S(r) = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{32\pi}{r}, \quad S'(r) = 4\pi r - \frac{32\pi}{r^2}.$$

$S' = 0$ gives $r^3 = 8$, so $r = 2 \text{ cm}$ and $h = 4 \text{ cm}$. $S'' > 0$ for $r > 0$, so this is a min. The best can has height equal to its diameter. ■

19. Antiderivatives.

- (c) A particle has $a(t) = 6t - 4$, $v(0) = 3$, $s(0) = 1$. Find $s(t)$.

Solution. $v(t) = 3t^2 - 4t + 3$ (the constant is $v(0)$), and $s(t) = t^3 - 2t^2 + 3t + 1$ (the constant is $s(0)$). ■

20. Riemann sums.

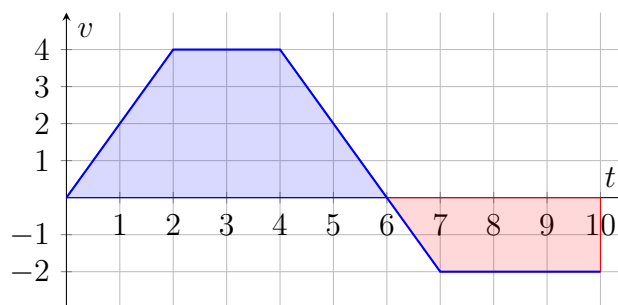
- (c) Write $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$ as a definite integral, then evaluate it.

Solution. Read $\Delta x = \frac{3}{n}$ and $x_i = 1 + \frac{3i}{n}$, so the interval starts at 1 and has length 3, and the height is $\sqrt{x_i}$.

$$\int_1^4 \sqrt{x} \, dx = \frac{2}{3} [x^{3/2}]_1^4 = \frac{2}{3}(8 - 1) = \frac{14}{3}.$$

21. The integral as area.

- (d) For the velocity graph below (v in m/s, t in s), find the displacement and total distance from $t = 0$ to $t = 10$, the position at $t = 10$ if $x(0) = 3 \text{ m}$, the acceleration at $t = 5$, and when the particle is farthest from its start.



Solution. Displacement is the signed area under v . Above the axis (blue): triangle 4, rectangle 8, triangle 4, total 16. Below (red): triangle 1 and rectangle 6, total 7.

$$\text{displacement} = 16 - 7 = 9 \text{ m}, \quad \text{distance} = 16 + 7 = 23 \text{ m}.$$

The position at $t = 10$ is $3 + 9 = 12$ m. The acceleration at $t = 5$ is the slope of the segment from $(4, 4)$ to $(7, -2)$, which is -2 m/s^2 .

The particle moves forward while $v > 0$ and backward after $t = 6$, so it is farthest from the start at $t = 6$, 16 m away. ■

22. Fundamental Theorem, Part 1.

(c) Find $g'(x)$ if $g(x) = \int_{2x}^{x^3} \frac{1}{1+t^4} dt$.

Solution. Split at 0 and apply FTC 1 with the chain rule to each piece.

$$g'(x) = \frac{3x^2}{1+x^{12}} - \frac{2}{1+16x^4}.$$

23. Fundamental Theorem, Part 2.

(c) What is wrong with $\int_{-1}^2 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^2 = -\frac{3}{2}$?

Solution. FTC 2 needs f continuous on $[a, b]$, and $\frac{1}{x^2}$ blows up at $x = 0$, inside $[-1, 2]$. The answer cannot be right anyway, because $\frac{1}{x^2} > 0$ and the integral of a positive function cannot be negative. (This integral is improper, and it diverges. Improper integrals come in Calculus 2.) ■

24. Substitution.

(c) Evaluate $\int_{-2}^1 x\sqrt{x+3} dx$.

Solution. Let $u = x + 3$, so $du = dx$, $x = u - 3$, and the bounds become 1 and 4.

$$\int_1^4 (u-3)u^{1/2} du = \left[\frac{2}{5}u^{5/2} - 2u^{3/2}\right]_1^4 = \left(\frac{64}{5} - 16\right) - \left(\frac{2}{5} - 2\right) = -\frac{8}{5}.$$

The answer is negative because $x\sqrt{x+3}$ is negative on most of $[-2, 1]$. ■

25. Area between curves.

- (b) Find the area between $y = \sqrt{x}$ and $y = \frac{x}{3}$, integrating with respect to x , then with respect to y .

Solution. They meet where $\sqrt{x} = \frac{x}{3}$: $x = 0$ and $x = 9$, at heights 0 and 3. The root is on top (at $x = 1$: $1 > \frac{1}{3}$).

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = \left[\frac{2}{3}x^{3/2} - \frac{x^2}{6} \right]_0^9 = 18 - \frac{27}{2} = \frac{9}{2}.$$

In y , solve each curve for x : the line is $x = 3y$ (on the right) and the root is $x = y^2$ (on the left).

$$\int_0^3 (3y - y^2) dy = \frac{27}{2} - 9 = \frac{9}{2}.$$

■

26. Average value.

- (b) Find the average value of $f(x) = x^2$ on $[0, 3]$ and a c in $[0, 3]$ with $f(c)$ equal to it.

Solution. $f_{\text{ave}} = \frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3} \cdot 9 = 3$. Then $c^2 = 3$ gives $c = \sqrt{3}$, which is in $[0, 3]$. The Mean Value Theorem for integrals says such a c always exists for continuous f . ■

27. Volumes.

- (c) Revolve the region between $y = \sqrt{x}$ and $y = x^2$ around the y -axis, and then around the line $x = -1$.

Solution. They meet at $(0, 0)$ and $(1, 1)$. Both axes are vertical, so slice horizontally. At height y the region runs from the parabola $x = y^2$ (inner) out to the root curve $x = \sqrt{y}$ (outer).

$$V_{y\text{-axis}} = \pi \int_0^1 (y - y^4) dy = \pi \left(\frac{1}{2} - \frac{1}{5} \right) = \frac{3\pi}{10}.$$

Around $x = -1$ each radius grows by 1: $R = \sqrt{y} + 1$ and $r = y^2 + 1$.

$$V = \pi \int_0^1 ((\sqrt{y} + 1)^2 - (y^2 + 1)^2) dy = \pi \int_0^1 (y + 2\sqrt{y} - y^4 - 2y^2) dy = \frac{29\pi}{30}.$$

■

28. Work.

- (c) A 10 m chain with mass 2 kg per meter hangs from the top of a building. How much work does it take to pull the whole chain up to the top? Use $g = 9.8 \text{ m/s}^2$.

Solution. After x meters have been pulled up, $10 - x$ meters still hang, with weight $2(9.8)(10 - x) = 19.6(10 - x)$ N. That is the force needed to lift the next small piece dx .

$$W = \int_0^{10} 19.6(10 - x) dx = 19.6 \left[10x - \frac{x^2}{2} \right]_0^{10} = 19.6 \cdot 50 = 980 \text{ J}.$$

Check: the chain's center of mass rises 5 m, and $(20 \text{ kg})(9.8)(5 \text{ m}) = 980 \text{ J}$. ■